



TARGET-90

JKBOSE CLASS-11TH

**LIMITS AND
DERIVATIVES**

One Shot

MATHS

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Topics *to be covered*

One Shot

LIMITS AND DERIVATIVES

LIMITS

The limit of a function $f(x)$ as x approaches a number a is the value that $f(x)$ gets closer to, and is denoted by $\lim_{x \rightarrow a} f(x)$.

$$\lim_{x \rightarrow a} f(x)$$

DERIVATIVES

The derivative of a function $f(x)$ with respect to x is the *rate of change* of $f(x)$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Some basic derivatives:

$$1 \quad \frac{d}{dx} (x^n) = nx^{n-1}$$

$$2 \quad \frac{d}{dx} (\sin x) = \cos x$$

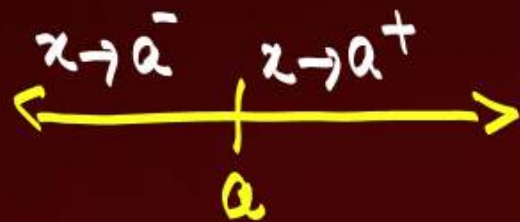
$$3 \quad \frac{d}{dx} (\cos x) = -\sin x$$

$$4 \quad \frac{d}{dx} (e^x) = e^x$$

$$5 \quad \frac{d}{dx} (\ln x) = \frac{1}{x}$$

$$6 \quad \frac{d}{dx} (a^x) = a^x \ln a$$

INTRODUCTION



★ **Limit of a function at a point a :** Suppose that f be any real valued function and $x=a$ be any real number. Then, a real number l is said to be the limit of the function at $x=a$ if $f(x)$ tends/approaches to l as x approaches to a .
i.e. $f(x) \rightarrow l$ as $x \rightarrow a$.

Symbolically, it's denoted as follows:

$$\lim_{x \rightarrow a} f(x) = l.$$

* If $f(x) \rightarrow l_1$ as $x \rightarrow a^+$ and $f(x) \rightarrow l_2$ as $x \rightarrow a^-$ where $l_1 \neq l_2$

i.e. $\lim_{x \rightarrow a^+} f(x) = l_1$ and $\lim_{x \rightarrow a^-} f(x) = l_2$, where $l_1 \neq l_2$.
(Right hand limit) left hand limit.

i.e. $\lim_{x \rightarrow a^+} f(x) \neq \lim_{x \rightarrow a^-} f(x)$, then $\lim_{x \rightarrow a} f(x)$ does not exist.

* $\lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a+h)$ (Right hand limit)

* $\lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} f(a-h)$ (left hand limit).

* If $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x)$, then $\lim_{x \rightarrow a} f(x)$ exists.

★ Methods for evaluating limits:

Direct Substitution:

* $f(x)$ don't change it's behaviour around $x=a$
i.e. the function is not being defined differently
in the right & left nbd. of a .

* Replace x by a provided the replacement
yields a finite number.

★ **Example:** Suppose that $f(x) = 2x^2 + 1 \quad \forall x \in \mathbb{R}$, then find
the limit of the function at $x=2$.

(Sol) $\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} (2x^2 + 1) = 2 \times 2^2 + 1 = 9$.

Factorisation Method: Suppose that $f(x) = \frac{x^2 - 25}{x - 5}$, $x \neq 5$
then find $\lim_{x \rightarrow 5} f(x)$?

(Sol) Here $\lim_{x \rightarrow 5} f(x) = \lim_{x \rightarrow 5} \left(\frac{x^2 - 25}{x - 5} \right)$ $\frac{0}{0}$ form.

$\Rightarrow \lim_{x \rightarrow 5} f(x) = \lim_{x \rightarrow 5} \frac{(x+5)(x-5)}{(x-5)} = \lim_{x \rightarrow 5} (x+5) = 10$.

EXAMPLE



Find the Limit of the following functions

(i) $f(x) = x^2 + 1$ at $x = 2$

(ii) $f(x) = 2x + 3$ at $x = 5$. H/w

EXAMPLE



Find the Limit of the following functions at the indicated points

(i) $f(x) = \frac{x^2 - 1}{x - 1}$, *HLW* $x \neq 1$ ($x = 1$)

(ii) $f(x) = \frac{x^2 - 5x + 6}{x - 3}$, $x \neq 3$ ($x = 3$)

$$\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \frac{(x-3)(x-2)}{(x-3)} = 3-2=1$$

QUESTION



What is the value of $\lim_{x \rightarrow 2} (x^2 + 3x - 4)$?

1 10 ✗

2 8 ✗

3 6 ✓

4 4 ✗

$$\begin{aligned} & \cancel{2^2} + 3 \times \cancel{2} - \cancel{4} \\ & = 6 \end{aligned}$$

QUESTION



Find the limit: $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$

1 ☒

2 ☒

3 ☒

4 Does not exist ☒

$$\begin{aligned} & \lim_{x \rightarrow 1} \left(\frac{x^2 - 1}{x - 1} \right) \\ &= \lim_{x \rightarrow 1} \frac{(x+1)(x-1)}{(x-1)} \\ &= \lim_{x \rightarrow 1} (x+1) = 1+1 = 2 \end{aligned}$$

QUESTION (NCERT EXAMPLER):

$$\lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x} \text{ is}$$

1

$n \checkmark$ (Sol) $\lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x} \left| \frac{0}{0} \text{ form} \right.$

2

$1 \times$ $= \lim_{x \rightarrow 0} \frac{1 + n_1 x + n_2 x^2 + \dots + n_n x^n - 1}{x}$

3

$-n \times$ $= \lim_{x \rightarrow 0} \frac{(n x + n_2 x^2 + n_3 x^3 + \dots + x^n)}{x}$

4

$0 \times$ $= \lim_{x \rightarrow 0} \frac{x (n + n_2 x + n_3 x^2 + \dots + x^{n-1})}{x} = n.$

QUESTION

If $\lim_{x \rightarrow a} f(x)$ exists, then $f(a)$ must exist

(A) True ✗

(B) False ✓

QUESTION

If $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$, then the limit at $x = a$ does not exist.

(A) True ✓

(B) False ✗

QUESTION (NCERT EXAMPLER):

$\lim_{x \rightarrow 0} \frac{|x|}{x}$ is equal to

1 $\times$

2 $\times$

3 $\times$

4 Does not exist ✓

$$\frac{|x|}{x} = \begin{cases} \frac{x}{x}, & x > 0 \\ -\frac{x}{x}, & x < 0 \end{cases}$$

$$\Rightarrow \frac{|x|}{x} = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases} = f(x)$$

$$\text{Here } \lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0} f(h) = 1$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{h \rightarrow 0} f(-h) = \lim_{h \rightarrow 0} (-1) = -1$$

$$\lim_{x \rightarrow 0^+} \frac{|x|}{x} \neq \lim_{x \rightarrow 0^-} \frac{|x|}{x}$$

$\Rightarrow \lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

Theorem – 1:



Let f and g be two functions such that both $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist.

Then

(i) Limit of sum of two functions is sum of the limits of the functions, i.e.,

$$\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

(ii) Limit of difference of two functions is difference of the limits of the function, i.e.,

$$\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

(iii) Limit of product of two functions is product of the limits of the functions, i.e.,

$$\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

(iv) Limit of quotient of two functions is quotient of the limits of the functions (whenever the denominator is non zero), i.e.,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$$

Theorem – 2:

For any positive integer n , $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$.



QUESTION (NCERT EXAMPLER):

$$\lim_{x \rightarrow 1} \frac{x^m - 1}{x^n - 1} \text{ is}$$

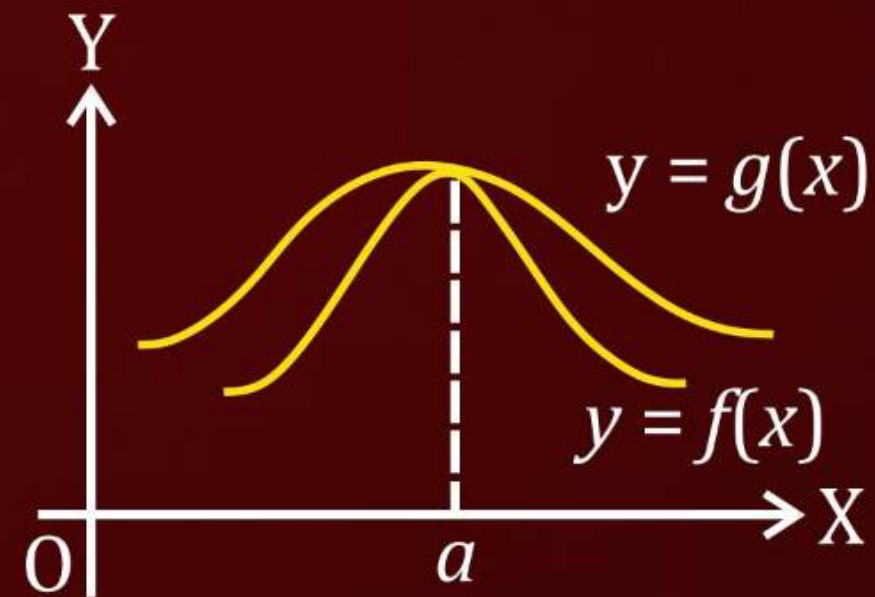
- 1 $\times$ $\lim_{x \rightarrow 1} \left(\frac{x^m - 1}{x^n - 1} \right) = \lim_{x \rightarrow 1} \left\{ \frac{\left(\frac{x^m - 1}{x - 1} \right)}{\left(\frac{x^n - 1}{x - 1} \right)} \right\}$
 - 2 $\frac{m}{n}$ ✓
 - 3 $-\frac{m}{n}$ $\times$
 - 4 $\frac{m^2}{n^2}$ $\times$
- $$= \frac{\lim_{x \rightarrow 1} \left(\frac{x^m - 1}{x - 1} \right)}{\lim_{x \rightarrow 1} \left(\frac{x^n - 1}{x - 1} \right)} = \frac{m(1)^{m-1}}{n(1)^{n-1}} = \frac{m}{n}$$

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n a^{n-1}$$

Theorem – 3:

Let f and g be two real valued functions with the same domain such that $f(x) \leq g(x)$ for all x in the domain of definition, For some a , if both $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist, then $\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x)$. This is illustrated in Figure.

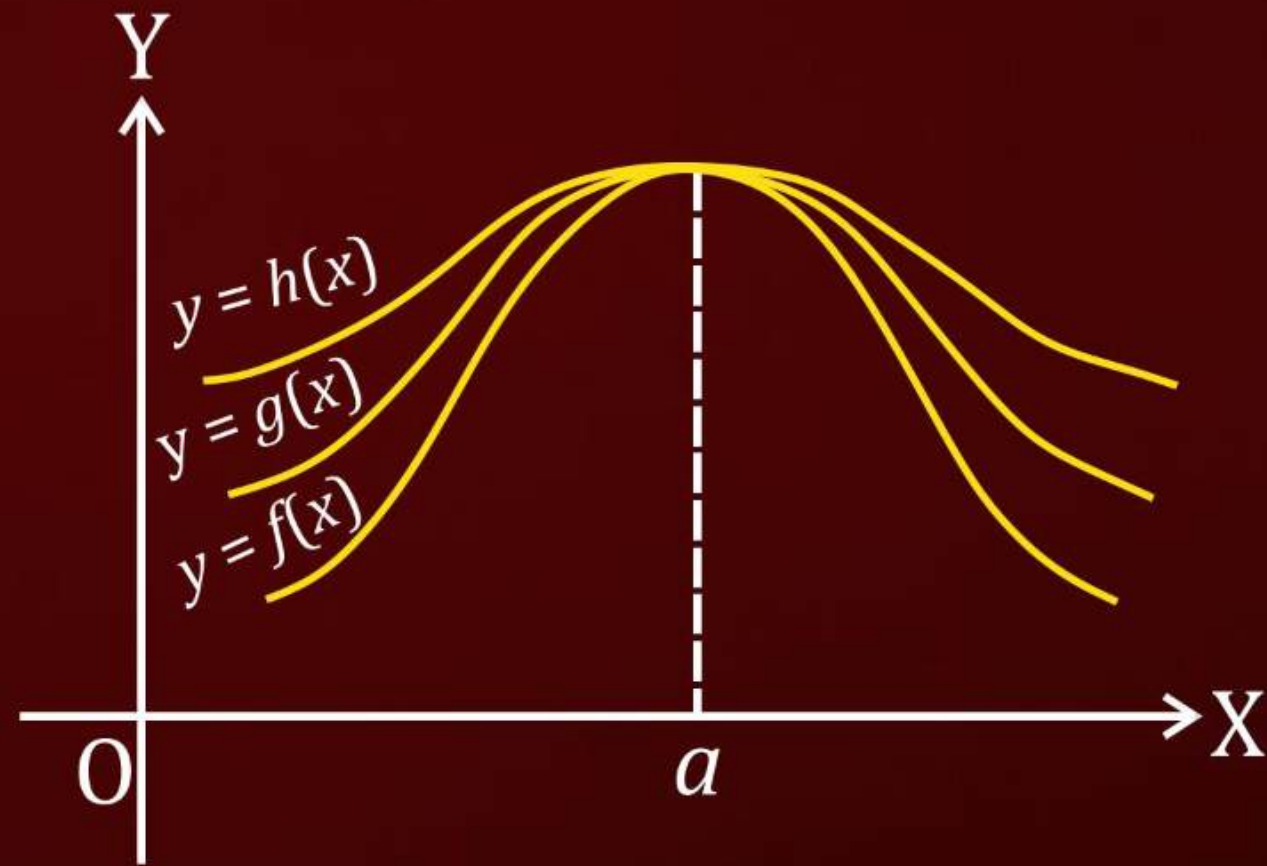
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Theorem – 4:

Let f, g and h be real functions such that $f(x) \leq g(x) \leq h(x)$ for all x in the common domain of definition. For some real number a , if $\lim_{x \rightarrow a} f(x) = l = \lim_{x \rightarrow a} h(x)$, and $\lim_{x \rightarrow a} g(x) = l$. This is illustrated in Figure.

Sandwich theorem
/ Squeeze Principle.



Limits of Trigonometric Functions

Theorem – 5:

The following are two important limits.

$$(i) \quad \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

$$(ii) \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$

$$(iii) \quad \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1.$$

QUESTION (NCERT EXAMPLER):

$\lim_{x \rightarrow 0} \frac{\sin x}{x(1 + \cos x)}$ is equal to

- 1 0 ✗
- 2 $1/2$ ✓
- 3 1 ✗
- 4 -1 ✗

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin x}{x(1 + \cos x)} &= \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \times \frac{1}{(1 + \cos x)} \right) \\ &= \lim_{x \rightarrow 0} \frac{\sin x}{x} \times \lim_{x \rightarrow 0} \frac{1}{(1 + \cos x)} \\ &= 1 \times \frac{1}{1+1} = \frac{1}{2} \end{aligned}$$

QUESTION (NCERT EXAMPLER):

$\lim_{x \rightarrow \pi/2} \frac{1 - \sin x}{\cos x}$ is equal to

1 0 ✓

2 -1 ✗

3 1 ✗

4 Does not exist ✗

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin x}{\cos x} \quad \left| \frac{0}{0} \text{ form} \right.$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{(1 - \sin x)(1 + \sin x)}{\cos x (1 + \sin x)}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{(1 - \sin^2 x)}{\cos x (1 + \sin x)} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cancel{\cos^2 x}}{\cancel{\cos x} (1 + \sin x)}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{(1 + \sin x)} = \frac{0}{1 + 1} = 0$$

QUESTION (NCERT EXAMPLER):

$\lim_{x \rightarrow 0} x \sin \frac{1}{x}$ is equal to

H/W

- 1 0
- 2 1
- 3 $1/2$
- 4 Does not exist

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 0} \frac{(x+1)^5 - 1}{x} \quad (\text{Pythagorean})$$

#

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 2} \frac{3x^2 - x - 10}{x^2 - 4} \quad \left| \frac{0}{0} \text{ form} \right.$$

$$= \lim_{x \rightarrow 2} \frac{3x^2 - 6x + 5x - 10}{x^2 - 2^2}$$

$$= \lim_{x \rightarrow 2} \frac{3x(x-2) + 5(x-2)}{(x+2)(x-2)}$$

$$= \lim_{x \rightarrow 2} \frac{(3x+5)(\cancel{x-2})}{(x+2)(\cancel{x-2})}$$

$$= \lim_{x \rightarrow 2} \left(\frac{3x+5}{x+2} \right) = \frac{3 \times 2 + 5}{2 + 2} = \frac{11}{4}$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 3} \frac{x^4 - 81}{2x^2 - 5x - 3} \quad (\text{pyq base})$$

QUESTION



Evaluate the following limits in Exercises 1 to 22.

$$\begin{aligned}\lim_{z \rightarrow 1} \frac{z^{\frac{1}{3}} - 1}{z^{\frac{1}{6}} - 1} &= \lim_{z \rightarrow 1} \frac{(z^{\frac{1}{6}})^2 - 1^2}{(z^{\frac{1}{6}} - 1)} \\ &= \lim_{z \rightarrow 1} \frac{(z^{\frac{1}{6}} - 1)(z^{\frac{1}{6}} + 1)}{(z^{\frac{1}{6}} - 1)} = 2\end{aligned}$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow \pi} \frac{\sin(\pi - x)}{\pi(\pi - x)} \quad \text{imp.}$$

QUESTION



Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 0} \frac{\cos 2x - 1}{\cos x - 1}$$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{2\cos^2 x - 1 - 1}{\cos x - 1} = \lim_{x \rightarrow 0} \frac{2\cos^2 x - 2}{\cos x - 1} \\ &= \lim_{x \rightarrow 0} \frac{2(\cancel{\cos x - 1})(\cos x + 1)}{\cancel{(\cos x - 1)}} = 2(2) = 4. \end{aligned}$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow 0} (\operatorname{cosec} x - \cot x) \quad \text{Imp}$$

QUESTION

Evaluate the following limits in Exercises 1 to 22.

$$\lim_{x \rightarrow \pi/2} \frac{\tan 2x}{x - \frac{\pi}{2}}$$

Imp

QUESTION



Find $\lim_{x \rightarrow 0} f(x)$, where $f(x) = \begin{cases} \frac{x}{|x|}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

H~.

QUESTION

$$\text{Suppose } f(x) = \begin{cases} a + bx, & x < 1 \\ 4, & x = 1 \\ b - ax, & x > 1 \end{cases}$$

and if $\lim_{x \rightarrow 1} f(x) = f(1)$ what are possible values of a and b ?

#

QUESTION



If the function $f(x)$ satisfies $\lim_{x \rightarrow 1} \frac{f(x) - 2}{x^2 - 1} = \pi$, evaluate $\lim_{x \rightarrow 1} f(x)$.

#

QUESTION

$$\text{If } f(x) = \begin{cases} mx^2 + n, & x < 0 \\ nx + m, & 0 \leq x \leq 1. \\ nx^3 + m, & x > 1 \end{cases}$$

For what value (s) of a does $\lim_{x \rightarrow 0} f(x)$ and $\lim_{x \rightarrow 1} f(x)$ exists?

$\neq$

Derivatives

★ **Definition:** Suppose that f be any real valued function and $x=a$ be any point in the domain of the function, then f is said to be differentiable at $x=a$ if $\lim_{x \rightarrow a} \frac{f(x)-f(a)}{x-a}$ exists finitely.

It's denoted by $f'(a)$. So

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x)-f(a)}{x-a} = \lim_{x \rightarrow a^+} \frac{f(x)-f(a)}{x-a}$$

$$\Rightarrow f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h} \quad \left(\begin{array}{l} \text{Derivative by first} \\ \text{principle or abinitio} \\ \text{Method} \end{array} \right)$$

★ **Derivative of $x^n, n \in \mathbb{Z}^+$:** Suppose that

$f(x) = x^n, n \in \mathbb{Z}^+$ be the given function.

Then by first principle, we get;

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{x^n + n x^{n-1} h + n_2 x^{n-2} h^2 + \dots + n_n h^n - x^n}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{n x^{n-1} h + n_2 x^{n-2} h^2 + \dots + n_n h^n}{h}$$

$$\Rightarrow f'(x) = n x^{n-1} \quad \text{or} \quad \frac{d}{dx}(x^n) = n x^{n-1}$$

$$\sin x - \sin y = 2 \cos\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right)$$

★ **Derivative of $\sin x$** : Suppose that $f(x) = \sin x$ be the given function. Then by first principle, we have;

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{2 \cos\left(\frac{x+h+x}{2}\right) \sin\left(\frac{x+h-x}{2}\right)}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{2 \cos\left(\frac{2x+h}{2}\right) \sin\left(\frac{h}{2}\right)}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \left(\frac{\cos\left(x+\frac{h}{2}\right) \sin\left(\frac{h}{2}\right)}{\frac{h}{2}} \right)$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \left\{ \cos\left(x+\frac{h}{2}\right) \times \frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)} \right\}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \cos\left(x+\frac{h}{2}\right) \times \lim_{h \rightarrow 0} \frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)}$$

$$\Rightarrow f'(x) = \cos x \times 1$$

$$\Rightarrow \boxed{f'(x) = \cos x} \text{ or } \frac{d}{dx}(\sin x) = \cos x$$

$$\cos x - \cos y = -2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right)$$

★ **Derivative of $\cos x$** : Suppose that $f(x) = \cos x$ be the given

function, then by first principle, we get;

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{-2 \sin\left(\frac{x+h+x}{2}\right) \sin\left(\frac{x+h-x}{2}\right)}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{-2 \sin\left(\frac{2x+h}{2}\right) \sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \left\{ \sin\left(x + \frac{h}{2}\right) \times \frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)} \right\}$$

$$\Rightarrow f'(x) = -\lim_{h \rightarrow 0} \sin\left(x + \frac{h}{2}\right) \times \lim_{h \rightarrow 0} \frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)}$$

$$\Rightarrow f'(x) = -\sin x \times 1$$

$$\Rightarrow \boxed{f'(x) = -\sin x} \text{ or } \frac{d}{dx}(\cos x) = -\sin x$$

★ **Derivative of Tanx**: Suppose that $f(x) = \tan x$ be the given

function. Then by first principle, we get;

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\tan(x+h) - \tan x}{h}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{1}{h} \left\{ \frac{\sin(x+h)}{\cos(x+h)} - \frac{\sin x}{\cos x} \right\}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{1}{h} \left\{ \frac{\sin(x+h)\cos x - \cos(x+h)\sin x}{\cos(x+h)\cos x} \right\}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{\sin(x+h-x)}{h \cos(x+h)\cos x}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \left\{ \frac{\sin h}{h} \times \frac{1}{\cos(x+h)\cos x} \right\}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{\sin h}{h} \times \lim_{h \rightarrow 0} \frac{1}{\cos(x+h)\cos x}$$

$$\Rightarrow f'(x) = 1 \times \frac{1}{\cos^2 x} = \sec^2 x$$

$$\Rightarrow \boxed{f'(x) = \sec^2 x} \text{ or } \frac{d}{dx}(\tan x) = \sec^2 x$$

EXAMPLE



Find the derivative of the function $f(x) = 2x^2 + 3x - 5$ at $x = -1$.

Also prove that $f'(0) + 3f'(-1) = 0$

$$f'(0) = 3.$$

$$f'(x) = \frac{d}{dx}(2x^2 + 3x - 5)$$

$$f'(0) + 3f'(-1) = 3 + 3 \times -1$$
$$= \cancel{3} - \cancel{3} = 0.$$

$$2 \times 2x + 3 \times 1$$
$$= 4x + 3.$$
$$f'(-1) = 4 \times -1 + 3 = -1.$$

Basic Rules of Differentiation:



$$* \frac{d}{dx}(kf(x)) = k \frac{d}{dx} f(x)$$

$$* \frac{d}{dx}(f(x) + g(x)) = \frac{d}{dx} f(x) + \frac{d}{dx} g(x)$$

$$* \frac{d}{dx}(fg) = f \frac{dg}{dx} + g \frac{df}{dx} \quad (\text{Product Rule})$$

$$* \frac{d}{dx}\left(\frac{f}{g}\right) = \frac{g \frac{df}{dx} - f \frac{dg}{dx}}{g^2} \quad (\text{Quotient Rule})$$

EXAMPLE

Find the derivative of $\frac{\cos x}{1 + \sin x}$

(Sol) Suppose that $f(x) = \frac{\cos x}{1 + \sin x}$ be the given

function. Then $f'(x) = \frac{(1 + \sin x) \frac{d(\cos x)}{dx} - \cos x \frac{d(1 + \sin x)}{dx}}{(1 + \sin x)^2}$

$$\Rightarrow f'(x) = \frac{-(1 + \sin x) \sin x - \cos x (0 + \cos x)}{(1 + \sin x)^2}$$

$$\Rightarrow f'(x) = \frac{-\sin x - \sin^2 x - \cos^2 x}{(1 + \sin x)^2}$$

$$\begin{aligned} &= -\sin x - 1 \\ &= -(1 + \sin x) \end{aligned}$$

$$\Rightarrow f'(x) = \frac{-\sin x - (\sin^2 x + \cos^2 x)}{(1 + \sin x)^2}$$

$$\Rightarrow f'(x) = \frac{-(\cancel{\sin x + 1})}{(\sin x + 1)^2} = \frac{-(\cancel{1 + \sin x})}{(1 + \sin x)^2}$$

$$\Rightarrow \boxed{f'(x) = \frac{-1}{1 + \sin x}} \text{ Hence the result.}$$

QUESTION

The derivative of the function $2x^2 + 6x + 1$ at $x = 3$ is:

- 1 18 ✓
- 2 17 ✗
- 3 16 ✗
- 4 12 ✗

$$\begin{aligned}
 &\checkmark \\
 &2 \times 2x + 6 + 0 \\
 &= 4x + 6 \quad \Bigg| \quad = 4 \times 3 + 6 \\
 &\quad \quad \quad \quad \quad \quad \quad \quad = 18. \\
 &\quad \quad \quad \quad \quad \quad \quad \quad x=3
 \end{aligned}$$

QUESTION (NCERT EXAMPLER):

If $f(x) = 1 + x + \frac{x^2}{2} + \dots + \frac{x^{100}}{100}$, then $f'(1)$ is equal to

1 $1/100$ ✗

2 100 ✓✓

3 Does not exist ✗

4 0 ✗

$$f'(x) = 0 + 1 + \frac{2x}{2} + \frac{3x^2}{2} + \dots + \frac{100x^{99}}{100}$$

$$\Rightarrow f'(x) = 1 + x + x^2 + \dots + x^{99}$$

$$\Rightarrow f'(1) = 1 + 1 + 1 + 1 + \dots + 1$$

$$\Rightarrow f'(1) = 100$$

QUESTION (NCERT EXAMPLER):

If $y = \frac{\sin(x + 9)}{\cos x}$ then $\frac{dy}{dx}$ at $x = 0$ is

h/w

1 $\cos 9$

2 $\sin 9$

3 0

4 1

QUESTION

The derivative of $\left(x + \frac{1}{x}\right)^3$ is:

1 $3x^2 + \frac{3}{x^4} + 3 + \frac{3}{x^2}$

2 $3x^2 - \frac{3}{x^4} + 3 - \frac{3}{x^2}$

3 $3x^2 + \frac{3}{x^4} + 3 - \frac{3}{x^2}$

4 None of these

$$x^3 + \frac{1}{x^3} + 3x \cdot \frac{1}{x} \left(x + \frac{1}{x}\right)$$

$$= x^3 + \frac{1}{x^3} + 3x + \frac{3}{x}$$

↓ Derivative

$$3x^2 - 3x^{-4} + 3 - 3x^{-2}$$

$$= 3x^2 - \frac{3}{x^4} + 3 - \frac{3}{x^2}$$

QUESTION



The derivative of the function $f(x) = (3x + 5)(1 + \tan x)$

1 ~~X~~ $3x \sec^2 x - 6 \sec^2 x - 3 + 3 \tan x$

2 ~~X~~ $3x \sec^2 x - 5 \sec^2 x - 3 + 3 \tan x$

3 ~~X~~ $3x \sec^2 x + \sec^2 x - 3 + 3 \tan x$

4  None of the above

$$\Rightarrow f(x) = (3x + 5)(1 + \tan x)$$

$$\Rightarrow f'(x) = (3x + 5) \frac{d}{dx}(1 + \tan x) + (1 + \tan x) \frac{d}{dx}(3x + 5)$$

$$\Rightarrow f'(x) = (3x + 5)(0 + \sec^2 x) + (1 + \tan x)(3 + 0)$$

$$\Rightarrow f'(x) = 3x \sec^2 x + 5 \sec^2 x + 3 + 3 \tan x$$

QUESTION



Find the derivative of the following functions from first principle.

(i) $x^3 - 27$

(ii) $(x - 1)(x - 2)$

(iii) $\frac{1}{x^2}$

(iv) $\frac{x + 1}{x - 1}$

$$\frac{(x-1) \times 1 - (x+1) \times 1}{(x-1)^2}$$

$$\frac{x-1-x-1}{(x-1)^2} = \frac{-2}{(x-1)^2}$$

(Sol) (iv) Suppose that $f(x) = \frac{x+1}{x-1}$ be the given function. Then by first principle, we get;

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \left\{ \frac{\frac{x+h+1}{x+h-1} - \frac{x+1}{x-1}}{h} \right\}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{(x+h+1)(x-1) - (x+1)(x+h-1)}{h(x+h-1)(x-1)}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{x^2 - x + x - h + x - 1 - x^2 - hx + x - x - h + 1}{h(x+h-1)(x-1)}$$

$$\Rightarrow f'(x) = \lim_{h \rightarrow 0} \frac{-2}{h(x+h-1)(x-1)}$$

$$\Rightarrow f'(x) = \frac{-2}{(x+0-1)(x-1)}$$

$$\Rightarrow f'(x) = \frac{-2}{(x-1)^2}$$

QUESTION



For the function

$$f(x) = \frac{x^{100}}{100} + \frac{x^{99}}{99} + \dots + \frac{x^2}{2} + x + 1. \text{ Prove that } f'(1) = 10 f'(0).$$

H/W

QUESTION



Find the derivative of

(i) $2x - \frac{3}{4}$

(ii) $(5x^3 + 3x - 1)(x - 1)$

(iii) $x^{-3}(5 + 3x)$ *h/w*

(iv) $x^5(3 - 6x^{-9})$

(v) $x^{-4}(3 - 4x^{-5})$

(vi) $\frac{2}{x+1} - \frac{x^2}{3x-1}$

QUESTION



Find the derivative of $\cos x$ from first principle.

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$$\begin{aligned} f(x) &= c. \quad f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ f(x+h) &= c. \quad , \quad f'(x) = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} (0) \\ &= 0. \end{aligned}$$

QUESTION



Find the derivative of

(i) $\sin x \cos x$

(ii) $\sec x$ H/W

(iii) $5 \sec x + 4 \cos x$

(iv) $\operatorname{cosec} x$

(v) $3 \cot x + 5 \operatorname{cosec} x$

(vi) $5 \sin x - 6 \cos x + 7$

(vii) $2 \tan x - 7 \sec x$

$$(\sec x)' = \sec x \tan x$$

$$(\cot x)' = -\operatorname{cosec}^2 x$$

$$(\operatorname{cosec} x)' = -\operatorname{cosec} x \cot x$$

Miscellaneous Examples:

Find the derivative of f from the first principle, where f is given by

(i) $f(x) = \frac{2x + 3}{x - 2}$ (PYQJKBose)

(ii) $f(x) = x + \frac{1}{2}$

Thank
You